The 1D even-monomial moment asymptotic

Tide retrospective, laplace seabed

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1 Setting

The laplace seabed's 2D separable-monomial track proves the mixed moment $\langle x^{2j_1} y^{2j_2} \rangle_t$ is asymptotically equivalent to a pure power of t (`KthKthMomentAsymptotic`), via a four-step pattern: exact factorisation into a product of 1D moments, rewriting the t-dependence as $C \cdot t^{-\cdots}$, a rescaled `Tendsto` to the constant, and the `IsEquivalent` packaging. That 2D result quietly leans on the corresponding one-dimensional statement, which existed nowhere as a standalone lemma. This tide supplies it.

2 The thing formalised

For the symmetric weight $\exp(-t\, x^{2k}/(2k)!)$ with $k \geq 1$, and every $t > 0$,

$$\langle x^{2j} \rangle_t = C(k,j) \cdot t^{-j/k}, \qquad C(k,j) := (2k)!^{\,j/k}\, \frac{\Gamma((2j+1)/(2k))}{\Gamma(1/(2k))},$$

and hence $\langle x^{2j} \rangle_t \sim C(k,j)\, t^{-j/k}$ at atTop. The Lean file gives all four rungs (`kthMomentConst`, `_eq_const_mul_rpow`, `_rescaled_tendsto`, `_isEquivalent_rpow`), the single-factor mirror of the 2D trio. The equivalence is, in fact, an eventual equality: the moment is exactly the power law for every positive t, so there is no asymptotic remainder to control — the content is entirely in the closed form the earlier tides already established.

3 Proof strategy

Each rung copies its 2D counterpart with one factor instead of two. The closed form $\langle x^{2j} \rangle_t = ((2k)!/t)^{j/k}\, \Gamma((2j+1)/(2k))/\Gamma(1/(2k))$ splits by `Real.div_rpow` into $(2k)!^{j/k}\, t^{-j/k}$ times the Gamma ratio, and `ring` finishes once $t^{-j/k}$ is written as $(t^{j/k})^{-1}$ by `Real.rpow_neg`. The rescaled `Tendsto` is the constant sequence in disguise: multiplying by $t^{j/k}$ and combining the powers with `Real.rpow_add` and `add_neg_cancel` collapses it to $C(k,j)$, so `tendsto_const_nhds.congr'` applies. The equivalence is `IsEquivalent.refl.congr_left` against the closed form, valid eventually via `eventually_gt_atTop`.

4 Roadblocks and resolutions

None; the tide compiled on the first attempt. The four-step pattern is documented in the seabed's own notes, and the single-factor specialisation is mechanical once the closest 2D file is read.

5 What was Mathlib, what was new

Mathlib supplied only the `Real.rpow` algebra and the `IsEquivalent`/`Tendsto` API. Everything else — the closed-form even moment and its integrability inputs — was already in the seabed. New here is the packaged 1D asymptotic itself, the base the 2D lift had been assuming. No GPT consult was needed.